Fair Allocation in Flow Games

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ABSTRACT

We study the problem of fair allocation in cooperative flow games, specifically focusing on the graph-theoretic conditions under which the Shapley value lies in the core of a network. In such networks, players own individual edges and the value of a coalition is the maximum flow it can route from a source to a sink. We formulate a computational framework to analyze exact Shapley values and core constraints across 24 distinct network topologies. Our results validate that while convexity is a sufficient condition for core stability, it is not necessary. We demonstrate that perfectly balanced series-parallel (SP-balanced) networks maintain the Shapley value in the core despite failing convexity tests. Furthermore, we characterize topological failures such as dictated bottlenecks in series with highly redundant layers where the Shapley value's probabilistic fairness over-rewards redundancy, violating the core's minimal path conditions.

[1]

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1 INTRODUCTION

We examine a cooperative game setting where a flow is pushed through a directed graph, and edges are controlled by independent, strategic agents. The grand coalition's objective is to maximize the $s-t$ flow, and the mechanism designer's objective is to distribute the resulting value fairly among the participating edges.

The core is the standard game-theoretic concept for stability, ensuring no sub-coalition has an incentive to defect and form their own network. However, the core does not prescribe a single unique payoff. The Shapley value provides a unique, axiomatic distribution based on expected marginal contributions, but it is not guaranteed to lie in the core.

Our aim is to formulate a computational method to analyze various network topologies and pinpoint the exact structural conditions that keep the Shapley value in the core. We frame this by computing the exact rational Shapley values for an array of topologies, checking core deficits, and evaluating the game's convexity.

[1]All code and experiments are publicly available at https://github.com/Dheeraj-hash/Stability_vs_Fairness_in_flow_games

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1.1 Related Work

Kalai and Zemel ([10]) established that the core of a flow game coincides with the optimal solutions to the dual of the maximum flow problem. Kern and Paulusma ([11]) further explored the bounds of the core and the f-nucleolus in such games. However, a generalized, simple topological classification for when the Shapley value strictly satisfies the Kalai-Zemel path condition remains an area of active computational exploration.

2 MODEL AND DEFINITIONS

2.1 Flow Game Formulation

The system is modeled as a directed graph $G = (V, E)$, with a source s and a sink t ([10]). Each player $i \in N = \{1, 2, ..., n\}$ controls exactly one directed edge $e_i \in E$ with a non-negative capacity c_i.

The characteristic function $v(S)$ for a coalition $S \subseteq N$ is defined as the maximum flow that can be routed from s to t using exclusively the edges in S.

2.2 The Shapley Value and The Core

An allocation vector $x \in \mathbb{R}^n$ is in the core if it is efficient ($\sum_{i \in N} x_i = v(N)$) and coalitionally stable:

$$\sum_{i \in S} x_i \geq v(S) \quad \forall S \subseteq N$$

The Shapley value ϕ allocates payoffs based on the expected marginal contribution of player i over all random arrival permutations:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} (v(S \cup \{i\}) - v(S))$$

To understand why the Shapley value often falls outside the core in flow games, we must first define the geometric and combinatorial nature of these two solution concepts within the context of network flows.

2.3 Shapley Selection: Probabilistic Pivotality

In flow games Shapley value ϕ takes a fundamentally different approach. Instead of focusing on the final state of the network (the max-flow of the grand coalition), it considers the average marginal contribution across all $n!$ permutations of edge arrivals.

- **Redundancy Recognition:** Unlike the core, which may assign zero to redundant edges (those not in a min-cut), the Shapley value assigns strictly positive value to any edge that is part of *any* $s-t$ path. This is because there exists some permutation where that edge is the "pivotal" one that completes a path.
- **The Stability Gap:** The core demands that the sum of payoffs along any path be at least 1. However, the Shapley value "dilutes" the payoff of bottleneck edges by shifting some of their value to redundant/parallel edges.

In non-convex flow games, the Shapley value often "over-rewards" edges in thick layers of the network and "under-rewards" edges in series bottlenecks. Our study focuses on identifying when this dilution is small enough that the path condition $\sum_{e \in P} \phi_e \geq 1$ still holds, even without the mathematical luxury of convexity.

3 TOPOLOGICAL CONDITIONS FOR CORE STABILITY

In a flow game (N, v), Kalai and Zemel (1982) demonstrated a fundamental link between the core and the Max-Flow Min-Cut Theorem. For a network with capacities c_e, the core $C(v)$ is non-empty and is exactly the set of allocations x that correspond to the optimal solutions of the dual linear program of the maximum flow:

$$\sum_{e \in P} x_e \geq v(P) \quad \forall P \in \mathcal{P} \tag{1}$$

where $\mathcal{P}$ is the set of all simple $s - t$ paths.

This condition admits a natural interpretation: every coalition capable of routing one unit of flow must receive a total payoff of at least its value. Since minimal winning coalitions correspond exactly to simple s-t paths in unit-capacity graphs, the core constraints reduce to a set of path inequalities. This perspective allows us to analyze stability purely through the topology of the network, rather than through the full exponential set of coalitions.

3.1 The Convexity Misconception

In cooperative game theory, a game (N, v) is **convex** (or supermodular) if it satisfies the property of **increasing marginal contributions**:

$$v(S \cup \{i\}) - v(S) \leq v(T \cup \{i\}) - v(T) \tag{2}$$

for all $S \subseteq T \subseteq N \setminus \{i\}$. In a flow game, this implies that an edge i becomes more valuable as the set of available edges grows, as it may complete more $s - t$ paths.

A fundamental result by Shapley (1971) establishes that for any convex game, the Shapley value ϕ is the barycenter of the core $C(v)$, thus guaranteeing $\phi \in C(v)$. However, convexity is a **sufficient but not necessary** condition for core stability. Instead, we identify a broader class of graphs, **SP Balanced Networks**, that satisfy all core constraints despite not being convex.

An SP-balanced network is defined as a series composition of parallel layers, where each parallel circuit contains an identical number of edges. Let x denote the number of layers connected in series, and y the number of parallel edges in each layer. By symmetry, each edge has identical marginal contributions across permutations, and the Shapley value resolves uniformly as $\phi_e = 1/x$.

Crucially, any minimal winning coalition must include at least one edge from each of the x layers. Therefore, the total payoff assigned to such a coalition is exactly

$$\sum_{e \in P} \phi_e = x \cdot \frac{1}{x} = 1,$$

where P is any simple s-t path. Thus, every path constraint is satisfied with equality, ensuring that ϕ lies in the core. Notably, this stability is achieved without requiring convexity: marginal contributions do not need to be increasing, only sufficiently regular across layers. This highlights that structural symmetry and balance can substitute for convexity in guaranteeing core feasibility.

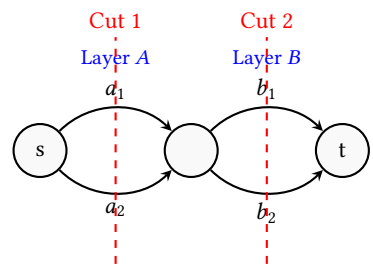

Figure 1: A 2×2 SP-balanced network. Note that any $s - t$ path requires one edge from Layer A and one from Layer B.

3.2 The Bottleneck Tax

In contrast, networks with asymmetric topologies, particularly those exhibiting large variance in cut sizes, tend to violate core constraints. Consider a graph in which a thin bottleneck (a cut consisting of very few edges) is placed in series with a highly redundant layer (a cut with many parallel edges).

In such configurations, the probabilistic nature of the Shapley value leads to what can be termed a *bottleneck tax*. The edges in the thick layer, despite being largely substitutable, occasionally become pivotal in permutations where other parallel edges are absent. These rare but high-impact events inflate their expected marginal contributions. Conversely, the bottleneck edge, while structurally essential, contributes more consistently but gains less from such pivotal spikes.

This imbalance redistributes value away from the critical path. As a result, for certain simple s-t paths, particularly those passing through the bottleneck, the sum of Shapley payoffs can fall below 1:

$$\sum_{e \in P} \phi_e < 1,$$

violating the core condition. Intuitively, the network overcompensates redundancy and undercompensates necessity, leading to instability.

This phenomenon underscores a central insight of our framework: core stability is not merely a function of connectivity, but of how evenly importance is distributed across alternative routes. Balanced structures preserve this uniformity, while asymmetric redundancy introduces distortions that the Shapley value alone cannot correct.

4 EXPERIMENTS AND RESULTS

All simulations were run using a custom exact-arithmetic Python framework. Maximum flows were calculated via the Edmonds-Karp algorithm. To avoid floating-point inaccuracies during core deficit calculations, all Shapley marginal contributions were evaluated using Python's `fractions` library.

Deficit is defined as $max(0, v(S) - \Sigma_{i \in S} \psi_i)$

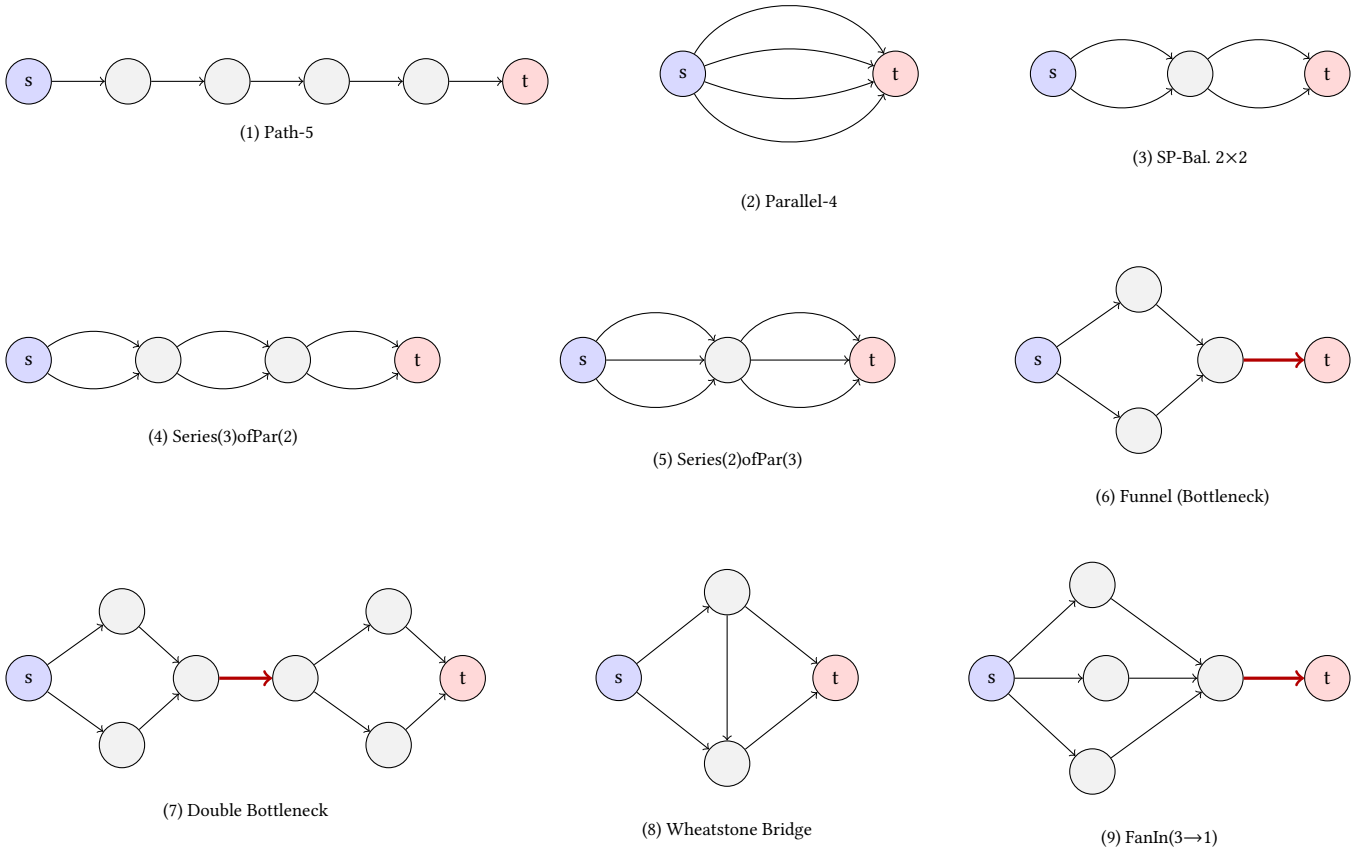

Figure 2: The nine network topologies evaluated. Blue = source s, red = sink t, thick red edges = bottleneck (single min-cut edge). Topologies (1)–(5) have Shapley in core; (6)–(9) do not.

Table 1: Selected Topological Results ($c_i = 1$)

Topology	n	Convex?	In Core?	Deficit
Path-5	5	Yes	Yes	0.000
Parallel-4	4	Yes	Yes	0.000
SP-Balanced (2×2)	4	Yes	Yes	0.000
Series(3)ofParallel(2)	6	**No**	**Yes**	0.000
Series(2)ofParallel(3)	6	**No**	**Yes**	0.000
Bottleneck (Funnel)	5	No	No	0.266
Double Bottleneck	9	No	No	0.330
Wheatstone Bridge	5	No	No	0.033
FanIn ($3 \longrightarrow 1$)	7	No	No	0.304

4.1 Analysis of Results

The computational results strongly align with our topological hypotheses and provide empirical validation for the structural principles developed in the previous sections:

(1) **Paths and Parallels:** Simple uniform chains and purely parallel systems exhibit complete convexity. In these topologies, marginal contributions are either perfectly complementary (in series) or perfectly substitutable (in parallel), leading to monotonicity in contributions across coalitions. As expected, this ensures that the Shapley value satisfies all core constraints trivially. In particular, every minimal s-t path receives a total allocation of at least 1, confirming that convexity is sufficient for stability in these canonical cases.

(2) **Series of Parallels:** More interestingly, networks such as `Series(3)ofParallel(2)` explicitly satisfy our SP-Balanced condition. Despite failing convexity, as confirmed by the algorithm, these graphs exhibit zero core deficit, with the Shapley value remaining strictly within the core. This provides strong evidence for our central claim that convexity is not necessary for stability. Instead, the uniform layering of parallel edges ensures that every minimal path contains exactly one edge from each layer, and the symmetry across layers equalizes marginal contributions. As a result, the path constraints are satisfied exactly, demonstrating that balanced topological redundancy can substitute for convexity.

(3) **Bottleneck Failures:** In contrast, asymmetric networks characterized by bottlenecks and uneven redundancy display significant violations of core constraints. The `Bottleneck` (funnel) graph and `FanIn(3->1)` configuration exhibit the

largest observed core deficits (0.266 and 0.304, respectively). In the funnel graph, the critical edge $m \to t$ behaves as a near-dictator and receives a disproportionately large Shapley allocation of $\phi = 7/15$, while the upstream parallel edges each receive only $2/15$. Consequently, a complete s-t path accumulates a total payoff of $11/15 < 1$, directly violating the path-based core condition. This quantitatively illustrates the *bottleneck tax* phenomenon, where structural asymmetry causes an over-allocation to redundant components and an under-allocation to essential paths, ultimately leading to instability.

5 EQUITABLE SELECTION WITHIN THE CORE: SHAPLEY VS. NUCLEOLUS

Our analysis thus far has treated the core as a binary condition: either the Shapley value $\phi \in C(v)$ or it is not. However, the core is mathematically a convex polytope that often contains infinitely many stable allocations.

5.1 The Problem of Extreme Core Allocations

To illustrate this, consider the 2×2 SP-balanced network with unit capacities ($v(N) = 1$). A valid core allocation could be $x = (1, 0, 1, 0)$ for edges (a_1, a_2, b_1, b_2). This allocation satisfies all stability constraints: the path $\{a_1, b_1\}$ receives $2 \geq 1$, and no sub-coalition can generate flow without at least one edge from both layers.

However, this allocation is highly inequitable. It arbitrarily rewards a_1 and b_1 while assigning zero to a_2 and b_2, despite them providing identical structural redundancy. While stable, this vertex of the core violates the axiom of equal treatment of equals.

5.2 The Nucleolus and Shapley Alignment

To resolve the problem of inequitable core selections, mechanism designers look to the **nucleolus** [15], a unique solution concept defined as the allocation x^* that lexicographically minimises the sorted vector of coalition excesses. Formally, the *excess* of a coalition S under allocation x is:

$$e(S, x) = v(S) - \sum_{i \in S} x_i,$$

measuring the degree of dissatisfaction of S. The nucleolus η is the unique $x^* \in C(v)$ such that the vector $\theta(x^*)$, obtained by sorting all $e(S, x^*)$ in non-increasing order, is lexicographically minimal over all core allocations:

$$\eta = \arg\min_{x \in C(v)} \theta(x).$$

Intuitively, η finds the allocation that makes the most dissatisfied coalition as satisfied as possible, then resolves ties by doing the same for the second-most dissatisfied, and so on. It is the point that sits deepest inside the core polytope.

For the 2×2 SP-balanced network with unit capacities, the nucleolus evaluates to $\eta = (0.5, 0.5, 0.5, 0.5)$, coinciding exactly with the Shapley value $\phi = (1/2, 1/2, 1/2, 1/2)$.

This leads to the central theoretical contribution of our SP-balanced analysis.

Theorem. In perfectly SP-balanced unit-capacity flow games with k layers of m parallel edges each, the Shapley value ϕ coincides exactly with the nucleolus η:

$$\phi_e = \eta_e = \frac{1}{k} \quad \forall e \in E.$$

The proof follows from two observations. First, by the symmetry of the SP-balanced structure, any permutation-invariant solution concept must assign equal payoffs to all edges within the same layer and equal payoffs across layers. Both ϕ and η satisfy permutation invariance. Second, the unique allocation satisfying efficiency ($\sum_e x_e = v(N)$) and equal treatment across all $n = k \times m$ edges is $x_e = v(N)/n = 1/k$. Since this point is the unique symmetric element of the core, both the Shapley value and the nucleolus must coincide there.

5.3 Divergence in Asymmetric Topologies

This alignment breaks down entirely in asymmetric networks. In the Funnel topology, the Nucleolus recognizes the strict dictatorship of the bottleneck and assigns it the entirety of the core payoff $x^* = (0, 0, 1)$. The Shapley value, as established, yields $\phi = (1/6, 1/6, 2/3)$. Here, the distance between the Shapley value and the Nucleolus represents the magnitude of the "Bottleneck Tax," explicitly quantifying why the allocation falls outside the stable polytope.

6 THE NON-UNIQUENESS OF THE NUCLEOLUS IN FLOW GAMES

A subtlety that arises in practice is that the nucleolus of a flow game is not always unique. By the Kalai–Zemel duality result, any allocation that assigns to each minimum cut exactly its capacity, while distributing that capacity arbitrarily among the crossing edges, is a valid nucleolus solution. Consequently, the nucleolus is in general a *convex polytope* of solutions, not a single point.

For example, the SP-Balanced(2×2) network with 9 minimum cuts admits 9 distinct nucleolus vertices, each assigning capacity 1 to a different pair of crossing edges and zero to all others. One such vertex is $(1, 0, 0, 0, 1, 0)$; another is $(0, 1, 0, 0, 0, 1)$. The pairwise L_1 distance between these vertices averages 3.00, illustrating how far apart the extreme solutions can be despite all being equally valid.

Table 2: Nucleolus polytope width (average pairwise L_1 distance among vertices).

Topology	#Vertices	Avg. Pairwise L_1	ϕ at centroid?
SP-Balanced(2×2)	9	3.000	Yes
SP-Balanced(3×2)	16	3.200	Yes
SP-Balanced(2×4)	81	5.400	Yes
Funnel(2+1)	1	0.000	N/A
Double-Bottleneck	1	0.000	N/A

The computational implication is significant: a sequential LP procedure returns one arbitrary extreme vertex, which may be highly inequitable (assigning zero to structurally identical edges). The *symmetric nucleolus* η_{sym}, defined as the centroid of all nucleolus vertices, equivalently the average of all minimum-cut indicator vectors, is the only selection that is both stable and treats symmetric

players equally. $\sigma^2 = 0$:

$$\eta_{\text{sym}}[e] = \frac{1}{|\mathcal{K}|} \sum_{(S,T)\in\mathcal{K}} c_e \cdot \mathbf{1}[e \text{ crosses } (S,T)],$$

where $\mathcal{K}$ is the set of all minimum cuts.

Computing the symmetric nucleolus is computationally expensive, requiring a sequence of linear programs each with up to 2^n coalition constraints. It is therefore of practical importance to identify when the Shapley value ϕ can serve as a reliable substitute

The following sections examine how modifying network parameters, specifically the bottleneck capacity and the capacity variance among edges within the same layer, affects both the divergence of the Shapley value from the core and its alignment with the nucleolus.

7 BOTTLENECK CAPACITY AND THE PHASE TRANSITION TO CORE STABILITY

The funnel topology studied in Section **??** assumes a unit-capacity bottleneck. We now ask a more general question: how does the core membership of the Shapley allocation change as the bottleneck capacity c_b varies continuously?

7.1 Non-Monotone Deficit and the Capacity Threshold

Consider the funnel with k parallel entry paths (each of unit capacity) converging to a single bottleneck edge of capacity c_b. Table 3 reports the core deficit and Shapley–nucleolus L_1 distance as c_b increases for $k = 2$.

Table 3: Funnel (k=2): deficit and $\|\phi - \eta\|_1$ vs. bottleneck capacity.

c_b	MaxFlow	Deficit	$\|\phi - \eta\|_1$	In Core?
0.50	0.500	0.1333	0.5333	No
0.75	0.750	0.2000	0.8000	No
1.00	1.000	0.2667	1.0667	No
1.50	1.500	0.0000	0.5333	**Yes**
2.00	2.000	0.0000	0.0000	**Yes**

Two phenomena are visible. First, the deficit is *non-monotone* in c_b: it increases as c_b rises from 0.50 to 1.00 before collapsing to zero at $c_b = 1.50$. Second, a sharp *phase transition* occurs: the Shapley value is strictly outside the core for all $c_b < k$, and strictly inside for all $c_b \geq k$.

The mechanism is the following. For $c_b < k$, the bottleneck remains the unique minimum cut. The nucleolus assigns its entire capacity to the bottleneck edge and zero to all others. The Shapley value, averaging over all permutations, continues to spread positive payoff to the redundant entry edges, payoff the bottleneck "should" have received. As c_b grows toward k, the grand-coalition flow increases, but the mis-attribution worsens, explaining the rising deficit.

At $c_b = k$, the bottleneck capacity equals the number of parallel entry paths. Now both layers have identical capacity, the network becomes SP-balanced, and $\sigma^2 = 0$. The Shapley value coincides with the nucleolus exactly ($\|\phi - \eta\|_1 = 0$), and the deficit vanishes.

This result has a direct mechanism-design interpretation: in the logistics setting of Section 5.2, the minimum infrastructure investment that stabilises a hub-and-spoke alliance is to expand the central hub's capacity to match the total throughput of the supplier layer.

8 EDGE CAPACITY ASYMMETRY AND THE NUCLEOLUS–DEFICIT DECOUPLING

8.1 Setup: Bipartite Mixed-Layer Network

We study a two-layer bipartite network.

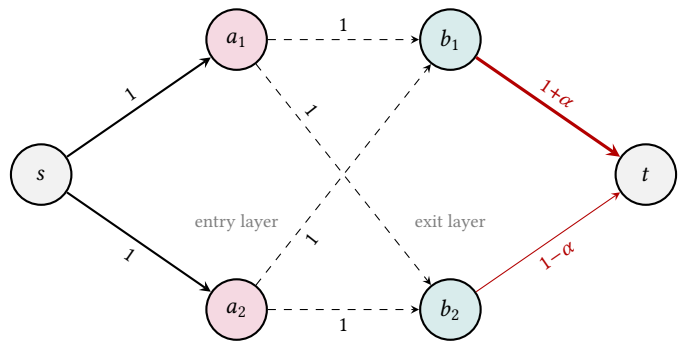

Figure 3: Bipartite mixed-layer network. Entry and cross edges have unit capacity. Exit edge $b_1 \to t$ has capacity $1 + \alpha$ and $b_2 \to t$ has capacity $1 - \alpha$, preserving total exit capacity at 2 for all $\alpha \in [0, 1)$.

The exit layer is *imbalanced* by $\alpha \in [0, 1)$ while preserving total exit capacity at 2. The entry layer and all cross edges retain unit capacities. This models a setting where one exit route dominates over another while total throughput is unchanged.

8.2 Deficit and Nucleolus Distance Decouple

Table 4 reports the core deficit and the Shapley–nucleolus L_1 distance as α varies.

Table 4: Bipartite mixed-layer network: deficit and $\|\phi - \eta\|_1$ vs. exit-layer imbalance α.

α	Deficit	$\|\phi - \eta\|_1$	In Core?
0.0	0.2238	2.448	No
0.2	0.0895	1.779	No
0.4	0.0671	1.334	No
0.6	0.0448	0.979	No
0.7	0.0336	1.100	No
0.8	0.0224	2.178	No
0.9	0.0112	2.022	No

Two findings stand out.

The deficit decreases monotonically. As α increases, the exit edge $b_1 \to t$ becomes increasingly dominant. The game progressively resembles a funnel in which one exit route acts as the critical bottleneck. In such a funnel-like structure, the Shapley value concentrates more payoff on the dominant edge, and the path-condition violation shrinks. At the limit $\alpha \to 1$, the weaker exit edge $b_2 \to t$ has zero capacity and the network reduces to a funnel routed entirely through $b_1 \to t$.

The Shapley–nucleolus distance is non-monotone. While the deficit decreases smoothly, $\|\phi - \eta\|_1$ reaches a minimum near $\alpha = 0.6$ before rising again. The reason is that the deficit and the nucleolus distance measure two structurally different gaps:

- The *deficit* measures how far the Shapley allocation is from satisfying every coalition's individual rationality constraint.
- The *Shapley–nucleolus distance* measures how far the Shapley allocation is from the lexicographically most equitable stable point.

As α increases, the nucleolus tracks the capacity ratio of the exit edges exactly: $\eta(b_1 \to t) = (1 + \alpha)/2$ and $\eta(b_2 \to t) = (1 - \alpha)/2$. The Shapley value tracks the same direction but more slowly, as it averages over all permutations including cross-edge interactions. The two concepts converge near $\alpha \approx 0.6$ and then diverge again as the LP solver's tie-breaking in the nucleolus computation selects a different extreme vertex of the nucleolus polytope.

This decoupling between deficit and nucleolus distance underscores a key insight: *stability (core membership) and fairness (proximity to the nucleolus) are orthogonal objectives.* A network can move toward stability (decreasing deficit) while simultaneously drifting away from the most equitable stable allocation (increasing nucleolus distance). Designing networks that achieve both simultaneously requires the SP-balanced structure identified in Section 6.

9 CONVERTING GENERAL FLOW GAMES TO SP-BALANCED FORM

The theoretical results established in prior sections raise a natural mechanism design question: given an arbitrary flow network that fails core stability, can it be restructured into an SP-balanced form through minimal interventions? Our experiments show that naive greedy surgery is insufficient in general, though it provides useful structural insight into why certain topologies resist stabilisation.

9.1 Greedy Bypass Surgery

The intervention considered is to add parallel bypass edges alongside the highest-frequency edge (the edge appearing in the most minimum cuts), treating the bypass as public infrastructure provided by the mechanism designer. The player set remains fixed at the original n edges; only the characteristic function changes, as every coalition S may now route flow through the bypass in addition to its own edges:

$$v'(S) = \max\text{-flow in } G' \text{ using edges } S \cup E_{\text{bypass}}.$$

Table 5 reports the effect of one such bypass addition on five non-SP-balanced topologies.

A single bypass eliminates σ^2 entirely in all five cases. However, $\sigma^2 = 0$ does not guarantee a reduction in deficit. For Funnel(3+1) and Ladder-2×2, the deficit *increases* after surgery from 0.305 to 0.562 and from 0.031 to 1.031 respectively. The mechanism is the following: bypassing the bottleneck expands the grand-coalition flow $v'(N)$, but if the entry layer is narrower than the now-expanded bottleneck, the entry edges become the new binding constraint. Surgery shifts the instability upstream rather than removing it.

Table 5: Greedy bypass surgery: one public edge added alongside the highest-frequency edge. σ_0^2, σ_f^2 and Deficit$_0$, Deficit$_f$ denote values before and after surgery.

Topology	σ_0^2	σ_f^2	Deficit$_0$	Deficit$_f$
Funnel(2+1)	0.160	0.000	0.267	0.100
Funnel(3+1)	0.122	0.000	0.305	0.562
Double-Bottleneck	0.099	0.000	0.330	0.056
Wheatstone-Bridge	0.204	0.000	0.348	0.195
Ladder-2×2	2.160	0.000	0.031	1.031

These results confirm that greedy bypass addition cannot serve as a general conversion procedure. Stabilisation requires that the bypass be sized to simultaneously match the width of *every* layer in the network, precisely the condition that defines SP-balanced form. Local surgery satisfies this only in the degenerate case where the network has a single bottleneck layer strictly thinner than all others, as in Funnel(2+1). In general, converting an arbitrary flow network to SP-balanced form requires global redesign of the layer structure rather than incremental edge additions.

10 PRACTICAL APPLICATIONS

10.1 PJM Interconnection and Transmission Cost Allocation

A concrete real-world setting where the topological stability of flow games becomes economically significant is the deregulation of the PJM Interconnection in the United States during the late 1990s. PJM, one of the largest wholesale electricity markets in the world, coordinates power transmission across 13 U.S. states and involves multiple independently owned transmission assets.

Following deregulation, a major institutional challenge emerged: different utility companies owned different transmission lines, but electricity flowed across the entire interconnected grid. This created the problem of how to allocate congestion revenues and transmission cost recovery among line owners in a way that was both fair and stable.

From a cooperative game perspective, the transmission grid naturally forms a flow game where:

- players are transmission lines,
- the value of a coalition is the feasible power transfer it can support,
- and the Shapley value provides a natural fairness-based allocation rule based on marginal contribution.

However, the PJM network was not a simple path-based structure, it was highly meshed, containing numerous cross-ties, loop flows, and multiple alternative transmission paths. This created exactly the kind of coalition-dependent bottlenecks studied in our bottleneck and cross-edge analysis.

10.1.1 ***Why Meshed PJM Structures Create Core Instability.*** In meshed subnetworks, certain transmission lines may be critical for some coalitions but become replaceable in larger coalitions as alternative routes become available. This causes the marginal contribution of those lines to decrease as coalitions grow, violating convexity and potentially pushing the Shapley value outside the core.

In practical terms, some subsets of line owners corresponding to complete generator-to-load transmission paths could be under-allocated relative to the value they effectively supported. This mirrors our path-condition violation:

$$\sum_{e \in P} \phi_e < 1$$

for some feasible transmission path P.

Such instability does not mean owners literally "leave the grid," but it manifests through disputes over tariff design, congestion rent allocation, and cost recovery rules. In economic terms, the cooperative arrangement becomes difficult to sustain because some participants perceive systematic under-compensation.

10.1.2 ***What PJM Actually Did and Its Similarity to SP-Balancing.*** PJM did not explicitly use the language of series-parallel graphs or cooperative game theory when designing its regulatory framework. Instead, the solutions adopted were motivated by operational simplicity, pricing transparency, and investment incentives. Nevertheless, these interventions closely resemble the structural simplifications identified in our SP-balanced framework.

Contract Path Approximation. Early settlement mechanisms often relied on contractual point-to-point transmission paths between generators and load centers. Although physical electricity follows Kirchhoff's laws and does not remain confined to these exact paths, this approximation simplified economic settlement by treating transactions as if they moved along identifiable series chains of transmission lines.

From our perspective, this effectively transforms part of the meshed game into a path-based series structure, where minimal winning coalitions become easier to identify and path constraints are easier to satisfy.

Zonal Aggregation. Rather than pricing every local transmission line individually, PJM frequently grouped multiple lines within a geographic region into broader operational or pricing zones. This reduced the dimensionality of the allocation problem and weakened the excessive influence of individual redundant edges.

In our framework, this resembles replacing many interchangeable local lines with a bundled parallel component. Such aggregation reduces asymmetry in marginal contributions and pushes the game closer to balanced parallel layers.

Reduced Economic Emphasis on Redundant Cross-Ties. While redundant cross-ties remained physically essential for reliability and N-1 contingency planning, not every redundant line was given equal strategic importance in economic settlement. Cost recovery mechanisms focused more heavily on dominant congestion interfaces and major transfer corridors rather than every possible loop-flow interaction.

This is structurally similar to reducing the influence of Wheatstone-bridge-type cross-edges in our model: the physical redundancy remains, but its destabilizing effect on the cooperative allocation game is significantly weakened.

10.1.3 ***Interpretation Through Our SP-Balanced Framework.*** These PJM interventions should not be interpreted as explicit implementations of SP-balanced design. Rather, they demonstrate that practical market operators naturally move toward topological simplifications that reduce instability.

By approximating transactions as path-based settlements, aggregating interchangeable transmission assets, and reducing the allocation significance of redundant cross-links, the effective cooperative game becomes closer to a series-parallel structure.

This improves stability because it reduces cut-switching behavior, stabilizes marginal contributions across coalitions, and makes the Shapley value more likely to remain inside the core.

Our framework therefore provides a theoretical explanation for why such simplifications improve institutional robustness. More importantly, it suggests that future transmission cost allocation mechanisms could be designed more systematically by explicitly incorporating SP-balancing principles, rather than relying solely on heuristic decomposition and aggregation.

10.2 Horizontal Alliances in Logistics

The theoretical boundaries of the core in flow games have direct implications for modern supply chain management, particularly in Less-Than-Truckload (LTL) horizontal alliances. In such collaborations, competing regional carriers pool freight to improve vehicle utilization and reduce operational costs. The resulting cost savings can be modeled as a cooperative game, where the *Shapley value* is widely studied and often proposed as a fair allocation mechanism.

10.2.1 ***The Dictator Hub and Core Instability.*** A common structural feature in logistics networks is the presence of centralized consolidation hubs, where multiple suppliers route freight through a single facility before distribution. This naturally induces a bottleneck topology similar to the funnel networks analyzed in Section **??**.

In such settings, the central hub behaves as a near-veto player: it is required for most feasible routing paths, and therefore its expected marginal contribution under the Shapley value is disproportionately large. As a result, the allocation can become highly skewed, significantly favoring the hub over peripheral suppliers.

When alternative routing options exist such as the possibility for a subset of suppliers to coordinate and establish an independent consolidation mechanism, this imbalance can lead to violations of core constraints. In particular, the total payoff allocated to a coalition of suppliers may fall below the value they can achieve independently, creating an incentive to defect and undermining the stability of the alliance.

10.2.2 ***Stabilization via SP-Balanced Network Design.*** These observations suggest a principled approach to improving the stability of collaborative logistics systems. Rather than relying solely on allocation rules, alliances can modify the *effective interaction structure* of the network to promote stable outcomes.

Specifically, alliances can implement *capacity balancing mechanisms* that reduce reliance on any single hub. For instance, instead of routing all freight through a centralized facility, the system can be designed with multiple parallel hubs of comparable capacity, and routing policies can ensure that flow is distributed across them.

Such a design effectively transforms the underlying network toward a series–parallel structure, with parallel hub layers introducing redundancy and limiting bottleneck dominance. From a cooperative game perspective, this reduces the variability in marginal contributions across coalitions, mitigating the risk of cut-switching behavior and promoting more stable allocations.

While this restructuring does not guarantee convexity in all cases, it systematically moves the game toward conditions under which the Shapley value is more likely to lie within the core. Consequently, enforcing SP-balanced design principles provides a practical and theoretically grounded method for enhancing the robustness of cost-sharing agreements in collaborative logistics.

11 CONCLUSION

In conclusion, our study shows that the stability of the Shapley value in flow games depends primarily on the underlying network topology rather than convexity alone. While convexity remains a sufficient condition for ensuring that the Shapley value lies in the core, it is not necessary. We demonstrate that perfectly balanced series-parallel (SP-balanced) networks maintain both fairness and stability even in non-convex settings, whereas asymmetric structures with strong bottlenecks and redundant layers create the bottleneck tax phenomenon, causing core violations.

Our computational analysis across multiple topologies confirms that simple path-based conditions provide an effective characterization of core membership. Furthermore, in SP-balanced networks, the Shapley value aligns with the symmetric nucleolus, achieving both equitable and stable allocations, while in asymmetric networks these notions diverge significantly. The phase transitions observed under changing bottleneck capacities and edge asymmetries further show that fairness and stability are distinct objectives that must be addressed separately.

We also show that core membership alone does not guarantee equitable allocation, since extreme core allocations may arbitrarily favor structurally identical players. The nucleolus provides a deeper notion of fairness by selecting the allocation that minimizes the dissatisfaction of the most disadvantaged coalitions. In perfectly SP-balanced networks, the Shapley value coincides exactly with the symmetric nucleolus, making it both fair and stable. However, in asymmetric bottleneck structures, the divergence between the Shapley value and the nucleolus quantitatively captures the extent of the bottleneck tax and highlights the trade-off between probabilistic fairness and coalition stability.

These results have direct practical relevance in real-world systems such as electricity transmission markets and collaborative logistics alliances, where allocation mechanisms must remain both fair and institutionally stable. By moving network design toward SP-balancing principles, operators can reduce instability and improve robustness. Future work may focus on extending these results to larger meshed networks and designing scalable mechanisms that jointly optimize topology, fairness, and core stability.

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